

120 Days of Quantitative Finance

Week 16: Stochastic Calculus + Time Series

Day 110: Monte Carlo Simulation of the Heston Model

1 Introduction

The Heston model is a stochastic volatility model where both asset prices and volatility evolve randomly over time.

It addresses limitations of constant volatility models by allowing volatility to be stochastic and mean-reverting.

2 Model Equations

The Heston model is defined as:

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}$$

$$dv_t = \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^{(2)}$$

$$\text{Corr}(dW_t^{(1)}, dW_t^{(2)}) = \rho$$

3 Interpretation

- S_t : asset price
- v_t : variance process

- κ : speed of mean reversion
- θ : long-term variance
- ξ : volatility of volatility
- ρ : correlation between price and volatility

4 Discretization (Euler Scheme)

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sqrt{v_t} S_t \sqrt{\Delta t} Z_1$$

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t)\Delta t + \xi \sqrt{v_t} \sqrt{\Delta t} Z_2$$

5 Correlated Brownian Motions

$$Z_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z'$$

where Z_1, Z' are independent standard normals.

6 Key Features

- stochastic volatility
- mean-reverting variance
- leverage effect (via ρ)

7 Applications

- option pricing
- volatility modeling
- risk management

8 Conclusion

Monte Carlo simulation allows us to numerically explore the behavior of stochastic volatility models like Heston, which are widely used in practice.